THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** do not touch (separate)	Compare 9 with 5+3=8 and |5-3|=2.
2	**(c)** 4	Midpoint of AB.
3	**(b)** 5	Diagonals of a rectangle are equal and bisect each other.
4	**(d)** 60	Triangle OAB is isosceles with OA = OB.
5	**(a)** 4	Subtract the radii.
6	**(d)** equal and perpendicular bisectors of each other	Property
7	**(a)** 2	Two pairs of equal opposite sides.
8	**(d)** 30	$2(l + b)$.
9	**(b)** touch externally	Compare 8 with 5+3=8 and |5-3|=2.
10	**(c)** 7	Midpoint of AB.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum of radii $= 10$, difference $= 2$ (1); The circles touch externally (1).	1+1
12	Diagonals are equal and bisect each other (1); OA = OB = 6 cm; BD = 12 cm (1).	1+1
13	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 3 cm from A (1).	1+1
14	Sum of radii $= 10$, difference $= 4$ (1); The circles touch internally (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw AB = 6 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3.0 cm from A (1).	1+1+1
16	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
17	Diameter $= 8$ cm (1); Longest chord $= 8$ cm (1); No: 9 cm exceeds the diameter (1).	1+1+1
	OR: Draw AB = 7 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3.5 cm from A (1).	1+1+1
18	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
	OR: Diameter $= 12$ cm (1); Longest chord $= 12$ cm (1); No: 13 cm exceeds the diameter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 8 cm (1); Draw perpendiculars at A and B and mark 6 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: AB = 7 (1); Sum of radii $= 9$; difference $= 1$ (1); They intersect at two points (1); Common points: 2 (2).	1+1+1+2
20	Draw AB = 12 cm (1); Draw perpendiculars at A and B and mark 5 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: AB = 9 (1); Sum of radii $= 9$; difference $= 3$ (1); They touch externally (1); Common points: 1 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 8	Diagonals of a rectangle are equal and bisect each other.
2	**(b)** 50	Triangle OAB is isosceles with OA = OB.
3	**(d)** 4	Subtract the radii.
4	**(b)** are equal but not perpendicular	Property
5	**(d)** 6	It equals the side.
6	**(c)** 44	$2(l + b)$.
7	**(c)** touch internally	Compare 4 with 6+2=8 and |6-2|=4.
8	**(b)** 9	Midpoint of AB.
9	**(c)** 10	Diagonals of a rectangle are equal and bisect each other.
10	**(b)** 40	Triangle OAB is isosceles with OA = OB.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diagonals are equal and bisect each other (1); OA = OB = 7 cm; BD = 14 cm (1).	1+1
12	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 5 cm from A (1).	1+1
13	Sum of radii $= 10$, difference $= 0$ (1); The circles intersect at two points (1).	1+1
14	Diagonals are equal and bisect each other (1); OA = OB = 9 cm; BD = 18 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw AB = 8 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 4.0 cm from A (1).	1+1+1
16	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
17	Diameter $= 16$ cm (1); Longest chord $= 16$ cm (1); No: 17 cm exceeds the diameter (1).	1+1+1
	OR: Draw AB = 9 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 4.5 cm from A (1).	1+1+1
18	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
	OR: Diameter $= 20$ cm (1); Longest chord $= 20$ cm (1); No: 21 cm exceeds the diameter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 6 cm (1); Draw perpendiculars at A and B and mark 4 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: AB = 4 (1); Sum of radii $= 9$; difference $= 5$ (1); They one lies inside the other (1); Common points: 0 (2).	1+1+1+2
20	Draw AB = 15 cm (1); Draw perpendiculars at A and B and mark 8 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: AB = 8 (1); Sum of radii $= 8$; difference $= 2$ (1); They touch externally (1); Common points: 1 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	Subtract the radii.
2	**(a)** perpendicular bisector of AB	Property
3	**(d)** 8	It equals the side.
4	**(a)** intersect at two points	Compare 5 with 4+4=8 and |4-4|=0.
5	**(c)** 12	Midpoint of AB.
6	**(b)** 13	Diagonals of a rectangle are equal and bisect each other.
7	**(b)** 30	Triangle OAB is isosceles with OA = OB.
8	**(d)** 4	Subtract the radii.
9	**(b)** 10	It equals the side.
10	**(b)** one lies inside the other	Compare 2 with 7+3=10 and |7-3|=4.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 8 cm from A (1).	1+1
12	Sum of radii $= 10$, difference $= 6$ (1); The circles do not touch (separate) (1).	1+1
13	Diagonals are equal and bisect each other (1); OA = OB = 11 cm; BD = 22 cm (1).	1+1
14	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 10 cm from A (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw AB = 10 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 5.0 cm from A (1).	1+1+1
16	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
17	Diameter $= 24$ cm (1); Longest chord $= 24$ cm (1); No: 25 cm exceeds the diameter (1).	1+1+1
	OR: Draw AB = 12 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 6.0 cm from A (1).	1+1+1
18	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
	OR: Diameter $= 30$ cm (1); Longest chord $= 30$ cm (1); No: 31 cm exceeds the diameter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 9 cm (1); Draw perpendiculars at A and B and mark 4 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: AB = 9 (1); Sum of radii $= 8$; difference $= 0$ (1); They do not touch (separate) (1); Common points: 1 (2).	1+1+1+2
20	Draw AB = 7 cm (1); Draw perpendiculars at A and B and mark 5 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: AB = 5 (1); Sum of radii $= 11$; difference $= 5$ (1); They touch internally (1); Common points: 1 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.